Classical and Recent Optimizers for Neural Network Training: A Controlled Study of Convergence, Stability, and Computational Cost

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Unconstrained Optimization — Week 12 — 17 November 2026

Abstract

(Member 4; each member supplies measured findings.) This project investigates when recent modifications of classical optimizers are useful for unconstrained neural network training under limited computational resources. We plan four paired comparisons: SGD versus DoG, momentum SGD versus Schedule-Free SGD, RMSProp versus momentum-free ADOPT, and Adam versus AdEMAMix. A shared protocol uses a small multilayer perceptron on Fashion-MNIST, a convolutional network on a fixed CIFAR-10 subset, and a character-level GRU on Tiny Shakespeare. Controlled diagnostics examine step-size adaptation, training-horizon dependence, second-moment normalization, and gradient-history length. The evaluation will distinguish optimization progress, predictive quality, variability, and computational cost. *[Replace planning language after experiments; add two measured findings and their limitations. No experimental results are claimed in this draft.]*

Code: *[Member 4: insert the real GitHub URL before submission.]*

1 Introduction and Related Work

(Member 1 integrates; Members 2–4 supply pair-specific literature notes.)

1.1 Motivation

Training a neural network requires minimizing an empirical loss over its parameters. Stochastic gradients make this feasible, but the trajectory depends on step sizes, gradient scaling, and the information retained from earlier updates [Bottou et al., 2018]. A comparison based only on final classification accuracy can miss slow optimization, unstable updates, or a large tuning cost. Our objective is therefore to connect differences in training behavior to specific algorithmic mechanisms.

A familiar dataset is useful when the experiment isolates a meaningful factor. For example, Zhang et al. [2024] investigate block heterogeneity using quadratic problems and neural networks, while Xie et al. [2025] examine the effect of coordinate geometry. These studies motivate controlled diagnostics alongside real tasks; they do not imply that classification difficulty alone measures poor conditioning.

1.2 Related work

We organize the literature around four modifications: dynamic distance-based step sizes [Ivgi et al., 2023], the relationship between scheduling and iterate averaging [Defazio et al., 2024], changes to adaptive normalization [Taniguchi et al., 2024], and the use of two gradient moving averages [Pagliardini et al., 2025]. Each modification addresses a different question, so each is paired with an appropriate classical baseline. Our study uses small networks and CPU budgets; it does not reproduce the scale of the original papers or claim that all four modifications solve the same problem.

1.3 Research questions and scope

We ask: (RQ1) Does each modification improve loss reduction or reduce tuning dependence under a shared budget? (RQ2) Does a targeted diagnostic support the proposed mechanism? (RQ3) Do observations persist across image classification and character prediction, and at what cost? Each member will independently implement and evaluate one pair. The planned contribution is a reproducible empirical comparison and an explanation of observed trade-offs, not a new optimizer. *[Member 1: update scope to match the completed experiments.]*

2 Problem Formulation and Methodology

2.1 Problem formulation

(Member 2; Member 3 verifies the data and model definitions.) For a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, let h_θ denote a network with parameters $\theta \in \mathbb{R}^d$. We consider

$$\min_{\theta\in\mathbb{R}^d} F(\theta), \qquad F(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell(h_\theta(x_i), y_i), \qquad g_t = \frac{1}{|B_t|}\sum_{i\in B_t} \nabla_\theta \ell(h_{\theta_t}(x_i), y_i). \tag{1}$$

There are no explicit constraints on θ. The neural network objectives are generally nonconvex. Image tasks use class cross-entropy; the language task averages next-character cross-entropy over predicted tokens. Weight decay is zero in the primary mechanism comparisons. The mini-batch B_t, parameter initialization, and training budget are matched within each pair.

For a controlled quadratic, we can use $q(\theta) = \frac{1}{2}(\theta - \theta^\star)^\top H(\theta - \theta^\star)$ with symmetric positive definite H and condition number $\kappa = \lambda_{\max}(H)/\lambda_{\min}(H)$. A stochastic oracle, if used, must be defined separately. The known optimum allows an objective gap on this problem; neural-network training loss is not a measured gap to a global optimum. *[Member 2: verify dimensions and the precise assumptions of each cited theorem.]*

2.2 SGD and DoG

(Member 1)

Plain SGD uses $\theta_{t+1} = \theta_t - \eta_t g_t$. Its learning rate must balance progress and stability. DoG instead constructs a scalar step size from distance traveled from the initial point and accumulated gradient norms [Ivgi et al., 2023]. We will test whether this reduces sensitivity to an externally selected learning rate. The paper's stochastic convex guarantee concerns a specified variant and its assumptions; it is not a guarantee for arbitrary nonconvex training. Conditioning diagnostics probe the limits of the mechanism rather than assuming that scalar adaptation removes anisotropy.

[Member 1: Write the exact DoG update and initialization, distinguish it from the theoretical variant, and specify the chosen distance floor. Compare with tuned SGD. Report the parameter-state count, tuning evaluations, and correctness checks.]

2.3 Momentum and Schedule-Free SGD

(Member 2)

For the convention used here, momentum SGD has $u_{t+1} = \mu u_t + g_t$ and $\theta_{t+1} = \theta_t - \eta_t u_{t+1}$, with zero initial velocity. Schedule-Free SGD combines interpolation and iterate averaging to avoid a decay schedule that requires a predetermined stopping horizon [Defazio et al., 2024]. The base

learning rate still requires selection. We will compare it with tuned momentum SGD using a documented cosine schedule and evaluate both early stopping and extended training. Evaluation must use the intended averaged point; the optimizer's train/eval state is distinct from the model's train/eval state.

[Member 2: Give all iterate definitions, averaging weights, initialization and train/eval transformations. Define the planned-horizon and oracle-horizon baselines. Check behavior against the official implementation and state the assumptions of the relevant theorem.]

2.4 RMSProp and momentum-free ADOPT

(Member 3)

A non-centered RMSProp baseline with no momentum maintains $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}$ and updates using coordinate-wise $g_t/(\sqrt{v_t}+\epsilon)$ [Hinton et al.]. We select ADOPT with $\beta_1 = 0$ to study a related normalization mechanism: excluding the current gradient from the second-moment estimate used to normalize that gradient [Taniguchi et al., 2024]. This is a declared setting of ADOPT, not a separately published RMSProp algorithm. The normalization convention and initialization differ and must be documented. Theoretical convergence under specified assumptions does not guarantee success for every neural-network setting or every learning rate.

[Member 3: Write the precise previous-moment update, first-step behavior, epsilon placement and clipping choice. Explain the pairing using Section 3.1 of ADOPT. Define the unconstrained stochastic diagnostic and distinguish a counterexample from a general guarantee.]

2.5 Adam and AdEMAMix

(Member 4)

Adam keeps moving averages of gradients and squared gradients, applies bias correction, and uses the resulting coordinate-wise update [Kingma and Ba, 2015]. AdEMAMix adds a slower gradient moving average to retain longer history alongside the fast moving average [Pagliardini et al., 2025]. We will test whether the additional history helps at the relatively short training horizons feasible on CPUs. The comparison uses zero weight decay for both methods. This isolates the history mechanism from differences between Adam and AdamW regularization. A no-slow-EMA ablation will use otherwise matched settings.

[Member 4: Give both complete updates, warm-up rules, beta values, mixing coefficient and bias corrections. Verify the ablation configuration rather than assuming it is exactly Adam. Measure optimizer-state bytes and explain short-horizon limitations.]

2.6 Mechanisms, assumptions, and implementation cost

(Member 2 integrates; each member completes their row.) All methods use first-order gradients, but retain different optimizer states and perform different scalar or coordinate-wise operations. Complexity should distinguish the gradient calculation from the optimizer update. Record actual persistent state tensors rather than only giving an $O(d)$ label.

Table 1: Method comparison and ownership. Complete cost and theorem details after implementation.

Pair	Main change	Assumptions and additional state	Writer
SGD / DoG	Dynamic scalar step	TBD	(Member 1)
SGDM / SF-SGD	Averaging and interpolation	TBD	(Member 2)
RMSProp / ADOPT-0	Previous-moment normalization	TBD	(Member 3)
Adam / AdEMAMix	Fast and slow gradient history	TBD	(Member 4)

Each implementation will be checked on a small deterministic problem or fixed gradient sequence against an independent calculation or trusted reference. Record discrepancies, version choices, numerical tolerances and any intentional deviations.

3 Experimental Setup

3.1 Datasets and preprocessing

(Member 3 maintains the shared specification; all members use the same manifest.) The planned tasks combine two image problems and one sequence problem (Table 2). Official image training data are split into training and validation partitions using saved indices; official test sets are reserved for final evaluation. The CIFAR-10 training subset will be stratified and selected only from the training partition. Text is split into contiguous training, validation and test regions before constructing next-character windows, preventing windows from crossing split boundaries. Preprocessing is fitted using training data only.

Table 2: Shared data and architecture proposal (Member 3 fills every row). Values are pilot starting points.

Task (writer)	Architecture proposal	Split and size to freeze
Fashion-MNIST (M3)	MLP: $784 \to 128 \to 64 \to 10$	Training/validation counts and index hash: TBD
CIFAR-10 (M3)	Two-block small CNN	About 10,000 training images; exact counts and index hash: TBD
Tiny Shakespeare (M3)	Character GRU, one layer, hidden size 64	Contiguous split, vocabulary, window length: TBD

Sources: Fashion-MNIST [Xiao et al., 2017], CIFAR-10 [Krizhevsky, 2009], and the Tiny Shakespeare text distributed with char-rnn [Karpathy]. MNIST may be used for implementation checks but is not a fourth primary task.

3.2 Model definitions

(Member 3; each member verifies the shared model fingerprint.) Within each task, both optimizers and all four members use the same architecture and loss. The initial proposal avoids batch normalization and dropout to simplify evaluation-point handling and reproducibility. Model widths, activation functions, initialization, vocabulary and parameter counts will be frozen after a short CPU pilot. *[Member 3: replace proposals with the exact architecture, parameter count and preprocessing configuration actually used.]*

3.3 Training and tuning protocol

(Member 1 integrates; each member records pair-specific configurations.) The proposed main experiment uses seeds 1, 2 and 3, paired initial parameters, and matched mini-batch orders. This gives 18 main runs per member and 72 across the team, excluding tuning and diagnostics. A separate tuning seed and validation set select method-specific settings. Equal learning rates are not required; comparable search budgets and documented ranges are. Data splits, optimizer hyperparameters, target-loss thresholds and update budgets must be frozen before the final test evaluation. No test curves are used for model selection.

A CPU pilot will set the common per-task budget. Each pair is timed on the same machine; absolute timings from different members' Macs will not be used to rank all eight optimizers. Weight decay is zero. Failed runs remain in the run manifest. A method that fails to reach a target is marked as unreached instead of receiving a zero time.

Table 3: Training protocol to complete before main experiments. Parentheses identify the writer.

Item	Frozen specification and writer
Splits and model hash	TBD (Member 3)
Per-task batch size / updates / evaluation interval	TBD (Member 1, with all members)
Seeds and tuning budget	Main: 1, 2, 3; tuning seed and trial count: TBD (Member 1)
SGD / DoG settings	TBD (Member 1)
SGDM / SF-SGD settings and schedules	TBD (Member 2)
RMSProp / ADOPT-0 settings	TBD (Member 3)
Adam / AdEMAMix settings	TBD (Member 4)
Target losses and timing boundary	TBD (Member 4, with all members)

3.4 Controlled diagnostics

(Member 2 integrates; each member supplies the exact design of their diagnostic.) The diagnostics test mechanisms rather than adding a larger benchmark. Each must state the varied factor, held-fixed conditions, measured quantity, and interpretation limits.

Table 4: Planned diagnostic factors; each row is completed by its named member.

Pair (writer)	Varied factor	Required control
SGD / DoG (M1)	Quadratic condition number and objective scale	Fix optimum, initialization and gradient oracle
SGDM / SF-SGD (M2)	Actual horizon relative to planned T	Declare schedules; include a horizon-aware tuned reference
RMSProp / ADOPT-0 (M3)	β_2 and specified gradient randomness	Same oracle draws; disclose clipping and epsilon
Adam / AdEMAMix (M4)	Training length and slow-EMA contribution	Match other hyperparameters; record state overhead

[Each member: freeze numerical factor levels and diagnostic budget in their own configuration.]

3.5 Metrics and computing environment

(Member 4 integrates; every member supplies their machine record.) Cross-entropy (CE) is measured using natural logarithms. Image tasks additionally report final test accuracy in percent; the text task reports held-out CE in nats per predicted character. Final training and validation losses are evaluated at the end of the fixed budget, at the appropriate evaluation point for each optimizer. Loss curves are plotted against updates and elapsed time. Report mean and sample standard deviation across seeds, the count of successful runs, and the number reaching the target. Three seeds support only a limited variability estimate.

Record elapsed training time including scheduled evaluations but excluding download and one-off dataset preparation; separately record optimizer-step time if profiling it. State storage in bytes must be distinguished from whole-process peak memory. Relative time ratios are computed only within a pair on one machine.

Table 5: Environment records; each member fills their row.

Writer	CPU / RAM / macOS	Python / libraries / threads	Environment record
(Member 1)	TBD	TBD	M1 environment.json
(Member 2)	TBD	TBD	M2 environment.json
(Member 3)	TBD	TBD	M3 environment.json
(Member 4)	TBD	TBD	M4 environment.json

4 Experimental Results and Analysis

4.1 SGD versus DoG

(Member 1; all cells, figures and analysis in this subsection.)

Hypothesis. Step-size adaptation will be assessed using the three common tasks and the pre-specified diagnostic. *[Member 1: state the falsifiable hypothesis before interpreting results.]*

Table 6: M1-MAIN: paired primary results (Member 1 fills all entries). Report mean ± sample SD and successful-run count out of 3. Test quality is accuracy (%) for Fashion/CIFAR and CE (nats/character) for Text. Train and validation columns are CE. No measured values yet.

Task	Optimizer (M1)	Train CE	Val CE	Test quality	Time (s)	$n/3$
Fashion	SGD	TBD	TBD	TBD	TBD	TBD
Fashion	DoG	TBD	TBD	TBD	TBD	TBD
CIFAR	SGD	TBD	TBD	TBD	TBD	TBD
CIFAR	DoG	TBD	TBD	TBD	TBD	TBD
Text	SGD	TBD	TBD	TBD	TBD	TBD
Text	DoG	TBD	TBD	TBD	TBD	TBD

Table 7: M1-DIAG: quadratic final objective gap, mean ± SD (Member 1). Replace the factor placeholders with frozen levels, expand rows as needed, and record tuning cost and reached counts in the companion CSV.

Factor / level (M1)	SGD gap	DoG gap	n
Condition number: [level]	TBD	TBD	TBD
Objective scale: [level]	TBD	TBD	TBD
Reference setting	TBD	TBD	TBD

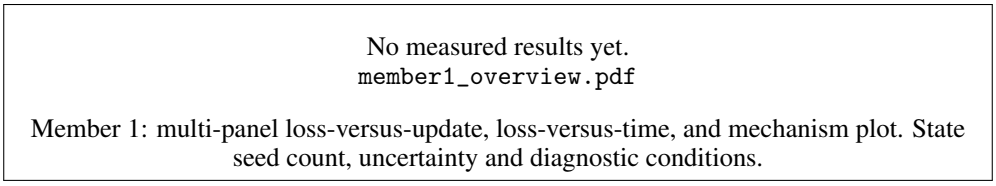


Figure 1: Member 1: multi-panel loss-versus-update, loss-versus-time, and mechanism plot. State seed count, uncertainty and diagnostic conditions.

Analysis and limitations. *[Member 1: quantify the paired difference, report failures, time-to-target and state storage from the accompanying CSVs, explain whether the diagnostic supports the mechanism, and identify an alternative explanation. Do not infer a universal ranking.]*

4.2 Momentum versus Schedule-Free SGD

(Member 2; all cells, figures and analysis in this subsection.)

Hypothesis. Training-horizon dependence will be assessed using the three common tasks and the pre-specified diagnostic. *[Member 2: state the falsifiable hypothesis before interpreting results.]*

Table 8: M2-MAIN: paired primary results (Member 2 fills all entries). Report mean ± sample SD and successful-run count out of 3. Test quality is accuracy (%) for Fashion/CIFAR and CE (nats/character) for Text. Train and validation columns are CE. No measured values yet.

Task	Optimizer (M2)	Train CE	Val CE	Test quality	Time (s)	$n/3$
Fashion	SGDM	TBD	TBD	TBD	TBD	TBD
Fashion	SF-SGD	TBD	TBD	TBD	TBD	TBD
CIFAR	SGDM	TBD	TBD	TBD	TBD	TBD
CIFAR	SF-SGD	TBD	TBD	TBD	TBD	TBD
Text	SGDM	TBD	TBD	TBD	TBD	TBD
Text	SF-SGD	TBD	TBD	TBD	TBD	TBD

Table 9: M2-DIAG: validation CE at the actual horizon (Member 2). "Planned" uses the original T schedule; "oracle" uses a schedule designed for the actual horizon. Define behavior after T. Report mean ± SD.

Actual horizon (M2)	SGDM planned	SGDM oracle	SF-SGD
$T/2$	TBD	TBD	TBD
T	TBD	TBD	TBD
$2T$	TBD	TBD	TBD

No measured results yet.
`member2_overview.pdf`

Member 2: multi-panel loss-versus-update, loss-versus-time, and mechanism plot. State seed count, uncertainty and diagnostic conditions.

Figure 2: Member 2: multi-panel loss-versus-update, loss-versus-time, and mechanism plot. State seed count, uncertainty and diagnostic conditions.

Analysis and limitations. *[Member 2: quantify the paired difference, report failures, time-to-target and state storage from the accompanying CSVs, explain whether the diagnostic supports the mechanism, and identify an alternative explanation. Do not infer a universal ranking.]*

4.3 RMSProp versus momentum-free ADOPT

(Member 3; all cells, figures and analysis in this subsection.)

Hypothesis. Normalization stability will be assessed using the three common tasks and the pre-specified diagnostic. *[Member 3: state the falsifiable hypothesis before interpreting results.]*

Table 10: M3-MAIN: paired primary results (Member 3 fills all entries). Report mean ± sample SD and successful-run count out of 3. Test quality is accuracy (%) for Fashion/CIFAR and CE (nats/character) for Text. Train and validation columns are CE. No measured values yet.

Task	Optimizer (M3)	Train CE	Val CE	Test quality	Time (s)	$n/3$
Fashion	RMSProp	TBD	TBD	TBD	TBD	TBD
Fashion	ADOPT-0	TBD	TBD	TBD	TBD	TBD
CIFAR	RMSProp	TBD	TBD	TBD	TBD	TBD
CIFAR	ADOPT-0	TBD	TBD	TBD	TBD	TBD
Text	RMSProp	TBD	TBD	TBD	TBD	TBD
Text	ADOPT-0	TBD	TBD	TBD	TBD	TBD

Table 11: M3-DIAG: stochastic normalization diagnostic (Member 3). Specify the objective, oracle and final metric (objective value or a known-optimum gap) before running. Report mean ± SD and failure counts.

β_2 / oracle (M3)	RMSProp metric	ADOPT-0 metric	Failures / runs
[Low value]	TBD	TBD	TBD
[Medium value]	TBD	TBD	TBD
[High value]	TBD	TBD	TBD

No measured results yet.
`member3_overview.pdf`

Member 3: multi-panel loss-versus-update, loss-versus-time, and mechanism plot. State seed count, uncertainty and diagnostic conditions.

Figure 3: Member 3: multi-panel loss-versus-update, loss-versus-time, and mechanism plot. State seed count, uncertainty and diagnostic conditions.

Analysis and limitations. *[Member 3: quantify the paired difference, report failures, time-to-target and state storage from the accompanying CSVs, explain whether the diagnostic supports the mechanism, and identify an alternative explanation. Do not infer a universal ranking.]*

4.4 Adam versus AdEMAMix

(Member 4; all cells, figures and analysis in this subsection.)

Hypothesis. Gradient-history length will be assessed using the three common tasks and the pre-specified diagnostic. *[Member 4: state the falsifiable hypothesis before interpreting results.]*

Table 12: M4-MAIN: paired primary results (Member 4 fills all entries). Report mean ± sample SD and successful-run count out of 3. Test quality is accuracy (%) for Fashion/CIFAR and CE (nats/character) for Text. Train and validation columns are CE. No measured values yet.

Task	Optimizer (M4)	Train CE	Val CE	Test quality	Time (s)	$n/3$
Fashion	Adam	TBD	TBD	TBD	TBD	TBD
Fashion	AdEMAMix	TBD	TBD	TBD	TBD	TBD
CIFAR	Adam	TBD	TBD	TBD	TBD	TBD
CIFAR	AdEMAMix	TBD	TBD	TBD	TBD	TBD
Text	Adam	TBD	TBD	TBD	TBD	TBD
Text	AdEMAMix	TBD	TBD	TBD	TBD	TBD

Table 13: M4-DIAG: validation CE and persistent optimizer-state cost (Member 4). The ablation removes the slow EMA while matching other settings; document whether it is equivalent to Adam. Report mean ± SD.

Condition (M4)	Adam	AdEMAMix	No slow EMA
Short horizon: CE	TBD	TBD	TBD
Longer horizon: CE	TBD	TBD	TBD
Optimizer state (bytes)	TBD	TBD	TBD

No measured results yet.
`member4_overview.pdf`

Member 4: multi-panel loss-versus-update, loss-versus-time, and mechanism plot. State seed count, uncertainty and diagnostic conditions.

Figure 4: Member 4: multi-panel loss-versus-update, loss-versus-time, and mechanism plot. State seed count, uncertainty and diagnostic conditions.

Analysis and limitations. *[Member 4: quantify the paired difference, report failures, time-to-target and state storage from the accompanying CSVs, explain whether the diagnostic supports the mechanism, and identify an alternative explanation. Do not infer a universal ranking.]*

4.5 Cross-task synthesis

(Member 1 integrates; Members 1–4 supply their own rows.)

Table 14: SYNTHESIS: evidence across tasks. Parentheses identify the writer. Do not pool raw timings across machines.

Pair (writer)	Where it helped	Where it did not	Mechanism / cost evidence
SGD / DoG (M1)	TBD	TBD	TBD
SGDM / SF-SGD (M2)	TBD	TBD	TBD
RMSProp / ADOPT-0 (M3)	TBD	TBD	TBD
Adam / AdEMAMix (M4)	TBD	TBD	TBD

[Member 1: answer RQ1–RQ3 using paired evidence and identify which observations do not transfer.]

5 Discussion and Conclusions

5.1 When do the modifications help?

(Member 3 integrates; each member supplies one evidence-supported claim.) *[Separate a reduction in tuning effort from a reduction in training loss or elapsed time. Identify settings where classical methods remain competitive. Connect each claim to a named table or panel, and explain discrepancies between synthetic diagnostics and neural-network tasks.]*

5.2 Limitations

(Member 2 integrates; all members report limitations encountered.) The design uses small networks, a limited CIFAR-10 training subset, and short CPU budgets. A character GRU cannot establish conclusions about large Transformer pretraining. Three main seeds and limited hyperparameter searches may miss small effects or better settings. Hardware differs between members, restricting time comparisons to matched pairs. Controlled quadratics isolate chosen factors but do not fully model nonconvex neural-network trajectories. *[Member 2: update these limitations to match actual runs and include observed failures.]*

5.3 Conclusions

(Member 4 integrates; all members approve claims about their pairs.) *[Answer each research question with measured evidence. State when a modification is worthwhile, its tuning or computational cost, and one concrete follow-up. Do not claim improvements before results exist.]*

Reproducibility statement

(Member 4 integrates; all members supply code, configurations and provenance.) *[Insert the real repository URL and commit, tested dependency versions, data preparation commands, frozen split manifest, run commands, exact figure/table regeneration commands, and observed runtime. The downloadable archive must include source, configurations, logs, aggregate data and report source; raw datasets and private files are excluded. Distinguish available code from planned interfaces.]*

Team contributions

(Member 1 integrates; each member fills their row.)

Table 15: Implementation and deliverable ownership; every member verifies their own contribution.

Writer	Independent implementation	Additional contribution
(Member 1)	SGD / DoG; verified code path: TBD	Introduction, tuning protocol, synthesis
(Member 2)	SGDM / SF-SGD; verified code path: TBD	Formulation, theory integration, limitations
(Member 3)	RMSProp / ADOPT-0; verified code path: TBD	Data/models specification, discussion
(Member 4)	Adam / AdEMAMix; verified code path: TBD	Metrics, abstract, conclusions, reproducibility

Reference maintenance. **(Member 1 integrates; Members 1–4 verify their sources.)**

References

Léon Bottou, Frank E. Curtis, and Jorge Nocedal. Optimization methods for large-scale machine learning. arXiv:1606.04838, version 3, 2018. URL https://arxiv.org/abs/1606.04838.

Aaron Defazio, Xingyu (Alice) Yang, Harsh Mehta, Konstantin Mishchenko, Ahmed Khaled, and Ashok Cutkosky. The road less scheduled. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://papers.nips.cc/paper_files/paper/2024/hash/136b9a13861308c8948cd308ccd02658-Abstract-Conference.html.

Geoffrey Hinton, Nitish Srivastava, and Kevin Swersky. Neural networks for machine learning: Lecture 6. Course slides, including the RMSProp method attributed to Tijmen Tieleman. URL https://www.cs.toronto.edu/~tijmen/csc321/slides/lecture_slides_lec6.pdf. Accessed 6 October 2026.

Maor Ivgi, Oliver Hinder, and Yair Carmon. DoG is SGD's best friend: A parameter-free dynamic step size schedule. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pages 14465–14499. PMLR, 2023. URL https://proceedings.mlr.press/v202/ivgi23a.html.

Andrej Karpathy. char-rnn: Tiny shakespeare data. GitHub repository. URL https://github.com/karpathy/char-rnn/tree/master/data/tinyshakespeare. Accessed 6 October 2026.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1412.6980.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. URL https://www.cs.toronto.edu/~kriz/cifar.html.

Matteo Pagliardini, Pierre Ablin, and David Grangier. The AdEMAMix optimizer: Better, faster, older. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/a2cf225ba392627529efef14dc857e22-Abstract-Conference.html.

Shohei Taniguchi, Keno Harada, Gouki Minegishi, Yuta Oshima, Seong Cheol Jeong, Go Nagahara, Tomoshi Iiyama, Masahiro Suzuki, Yusuke Iwasawa, and Yutaka Matsuo. ADOPT: Modified Adam can converge with any β_2 with the optimal rate. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/84d286e32bbee8fa3a86ee9c50e00081-Abstract-Conference.html.

Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. arXiv:1708.07747, 2017. URL https://arxiv.org/abs/1708.07747.

Shuo Xie, Mohamad Amin Mohamadi, and Zhiyuan Li. Adam exploits ℓ_∞-geometry of loss landscape via coordinate-wise adaptivity. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/1700ad4e6252e8f2955909f96367b34d-Paper-Conference.pdf.

Yushun Zhang, Congliang Chen, Tian Ding, Ziniu Li, Ruoyu Sun, and Zhi-Quan Luo. Why transformers need Adam: A hessian perspective. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/ee0e45ff4de76cbfdf07015a7839f339-Paper-Conference.pdf.